

TITAN research-software demonstration

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Research-software output: internally derived TCGA estimates

COAD - TCGA-AA-A01F

UNVALIDATED RESEARCH OUTPUT — BINARY TCGA OOF SCORE RANK (NOT PROBABILITY) — NO INDEPENDENT EXTERNAL VALIDATION — NOT FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION

Models applied	Continuous	Binary	Slides pooled
19	10	9	1

This research-software demonstration applies models developed and evaluated only in TCGA. Every displayed value is an **internally derived TCGA estimate**, not external performance evidence or a diagnostic output. Binary LDA scores are uncalibrated. Every displayed **TCGA OOF score rank (not probability)** describes relative position only.

By default, TITANPred excludes binary models with fewer than 50 participants in either development class. Such models appear only after the caller explicitly sets `include_limited_evidence = TRUE`.

Prominent tissue-source-site sensitivity warning

12 model(s) in this output fell below their original effect threshold under TCGA tissue-source-site-grouped internal validation. These calls must not be highlighted without this warning. Tissue-source-site grouping is not institutional or scanner-level external validation.

Endpoint (type)	TSS-grouped metric	Grouped minus random	Warning category
APC (binary)	BA 0.543	-0.250	Near chance or worse
PI3K (binary)	BA 0.593	-0.020	Below effect threshold
Aneuploidy score (continuous)	Q2 0.173	-0.086	Below effect threshold
Deleted arm count (continuous)	Q2 0.147	-0.066	Below effect threshold
MANTIS score (continuous)	Q2 0.113	-0.104	Below effect threshold
MSIsensor score (continuous)	Q2 0.026	-0.229	Below effect threshold
Aneuploidy Score (continuous)	Q2 0.174	-0.096	Below effect threshold
Lymphocyte Infiltration Signature Score (continuous)	Q2 0.048	-0.238	Below effect threshold
Nonsilent Mutation Rate (continuous)	Q2 0.088	-0.127	Below effect threshold
SNV Neoantigens (continuous)	Q2 -0.013	-0.252	Near chance or worse
Silent Mutation Rate (continuous)	Q2 -0.003	-0.236	Near chance or worse
TIL Regional Fraction (continuous)	Q2 0.176	-0.278	Below effect threshold

Case context (not used as model input)

Shared selection features. Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins. Nodal status was not matched: example A was pN1, whereas the available slide summary for example B did not state a nodal category.

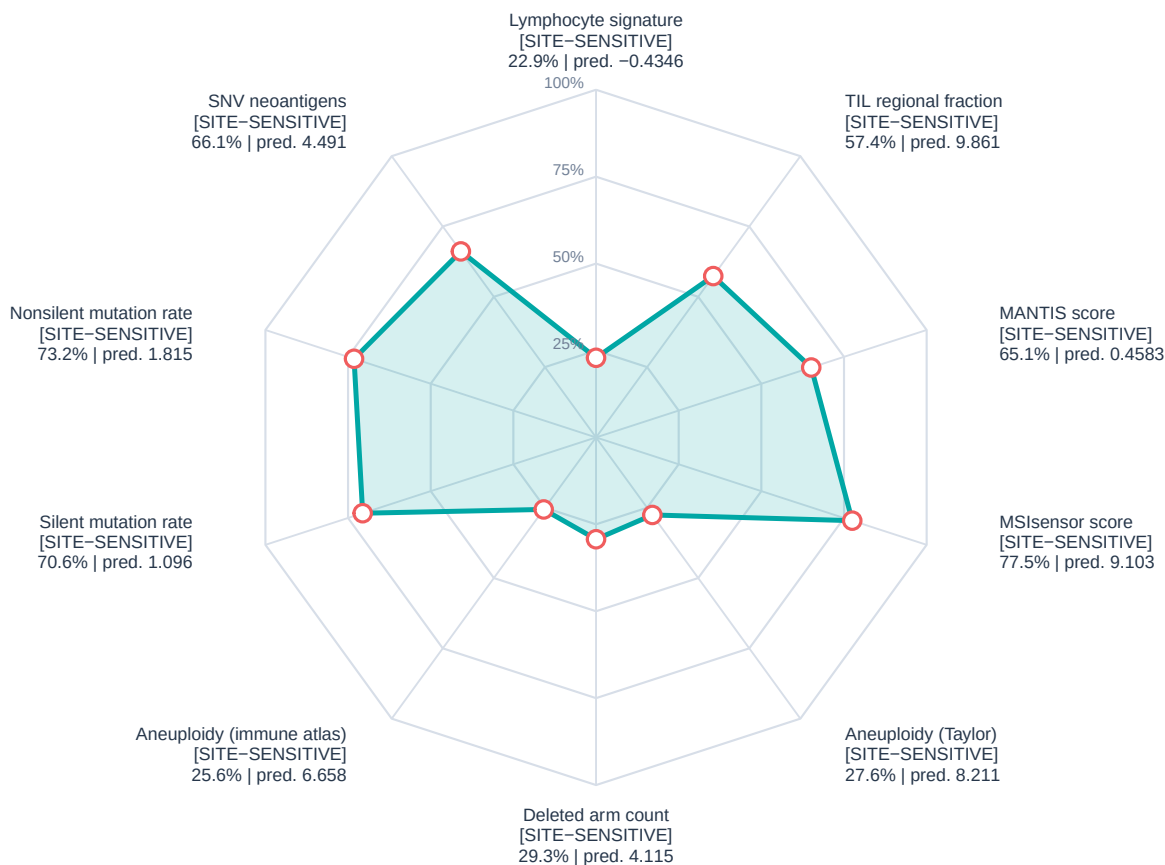
Pathology summary. The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

Clinical-context source. Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

The pathology field is descriptive context only; it was not a predictor, training target, or validation outcome for TITANPred. Treatment and response narratives are intentionally excluded from this research-software output.

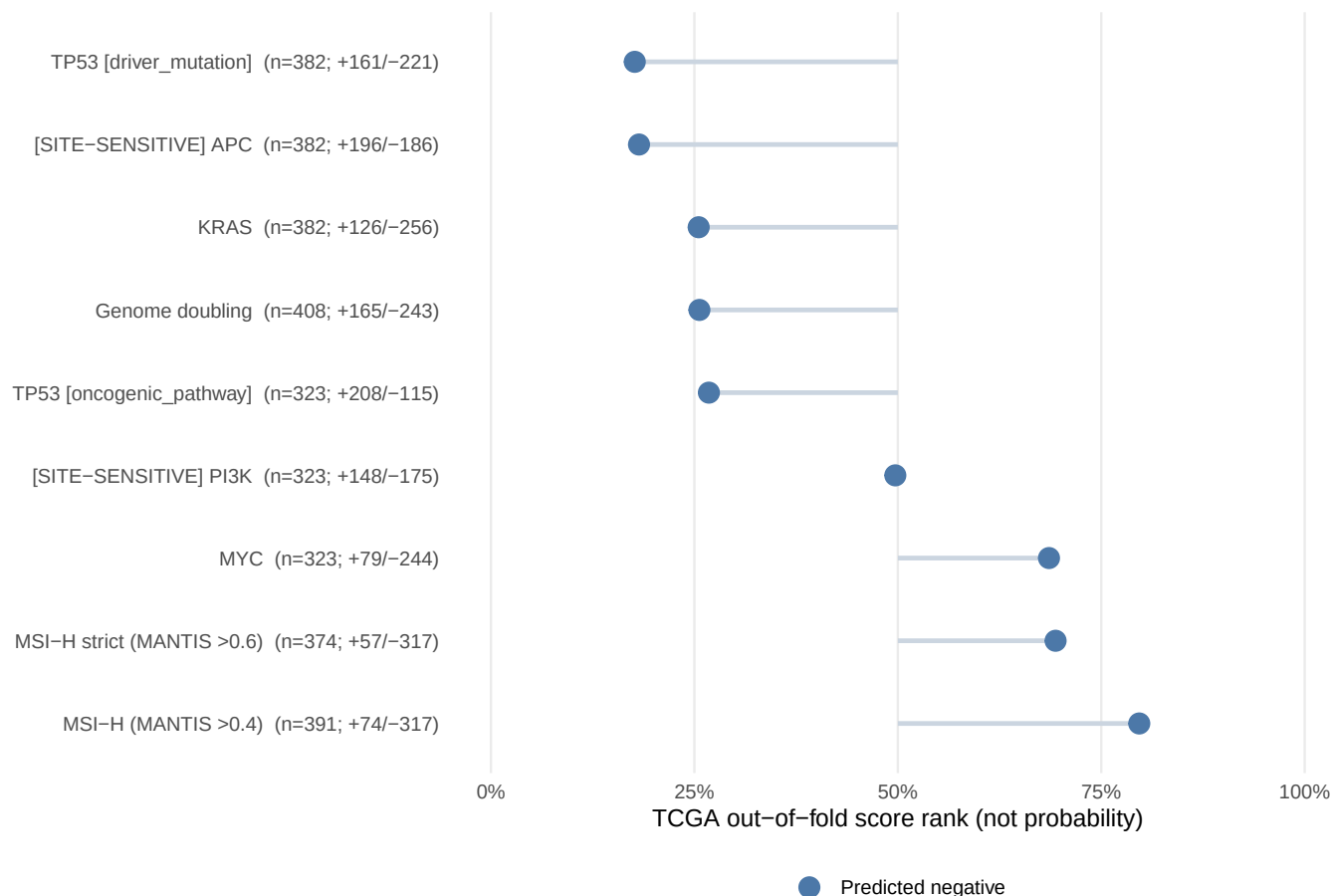
Continuous internally derived TCGA estimates

[SITE-SENSITIVE] marks models below the original effect threshold under TCGA TSS grouping



Binary research-model calls – rank is not probability

TCGA out-of-fold LDA-score rank (not probability); [SITE-SENSITIVE] models req



Continuous internally derived estimates

Family	Endpoint	Prediction	TCGA OOF prediction percentile (not prob)
microsatellite_instability	MSIsensor score	9.103	77.5%
thorsson	Lymphocyte Infiltration Signature Score	-0.4346	22.9%
thorsson	Aneuploidy Score	6.658	25.6%
thorsson	Nonsilent Mutation Rate	1.815	73.2%
aneuploidy	Aneuploidy score	8.211	27.6%
aneuploidy	Deleted arm count	4.115	29.3%
thorsson	Silent Mutation Rate	1.096	70.6%
thorsson	SNV Neoantigens	4.491	66.1%
microsatellite_instability	MANTIS score	0.4583	65.1%
thorsson	TIL Regional Fraction	9.861	57.4%

Binary research-model calls

Calls and development denominators.

Endpoint	Class	LDA score	TCGA OOF score rank (not probability)	Training n (+/-)
Genome doubling	0	-2.153	25.6%	408 (+165/-243)
APC	0	-2.982	18.2%	382 (+196/-186)
KRAS	0	-3.159	25.5%	382 (+126/-256)
TP53 [driver_mutation]	0	-3.171	17.7%	382 (+161/-221)
MSI-H (MANTIS >0.4)	0	-1.165	79.7%	391 (+74/-317)
MSI-H strict (MANTIS >0.6)	0	-4.352	69.4%	374 (+57/-317)
MYC	0	-1.023	68.6%	323 (+79/-244)
PI3K	0	-0.2634	49.7%	323 (+148/-175)
TP53 [oncogenic_pathway]	0	-0.05124	26.8%	323 (+208/-115)

Held-out repeated-CV evidence. PPV and NPV use the observed TCGA prevalence.

Endpoint	BA	AUROC	PR-AUC	PPV	NPV	Evidence
Genome doubling	0.692	0.760	0.678	0.636	0.750	>=50 per class
APC	0.794	0.848	0.814	0.760	0.851	>=50 per class
KRAS	0.692	0.781	0.581	0.600	0.794	>=50 per class
TP53 [driver_mutation]	0.778	0.854	0.793	0.718	0.828	>=50 per class
MSI-H (MANTIS >0.4)	0.732	0.834	0.634	0.677	0.894	>=50 per class
MSI-H strict (MANTIS >0.6)	0.782	0.915	0.741	0.750	0.931	>=50 per class
MYC	0.605	0.676	0.417	0.515	0.801	>=50 per class
PI3K	0.607	0.640	0.583	0.582	0.635	>=50 per class
TP53 [oncogenic_pathway]	0.668	0.734	0.815	0.752	0.624	>=50 per class

Binary development reliability

Fold class counts and component selection.

Endpoint	Minimum outer test (+/-)	Minimum inner training (+/-)	Components: median (range)
Genome doubling	+33 / -48	+105 / -155	6.0 (4-10)
APC	+39 / -37	+124 / -118	2.0 (1-4)
KRAS	+25 / -51	+80 / -163	4.0 (2-10)
TP53 [driver_mutation]	+32 / -44	+102 / -140	2.0 (2-7)
MSI-H (MANTIS >0.4)	+14 / -63	+47 / -202	6.0 (4-9)
MSI-H strict (MANTIS >0.6)	+11 / -63	+36 / -202	7.0 (3-10)
MYC	+15 / -48	+50 / -156	4.0 (2-5)
PI3K	+29 / -35	+94 / -112	4.0 (2-10)
TP53 [oncogenic_pathway]	+41 / -23	+132 / -73	3.0 (1-6)

Repeat-to-repeat held-out prediction stability.

Endpoint	Score Spearman rho	Class agreement
Genome doubling	0.850	0.857
APC	0.945	0.944
KRAS	0.825	0.858
TP53 [driver_mutation]	0.948	0.931
MSI-H (MANTIS >0.4)	0.856	0.940
MSI-H strict (MANTIS >0.6)	0.804	0.948
MYC	0.861	0.905
PI3K	0.721	0.787
TP53 [oncogenic_pathway]	0.853	0.895

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.3%.
- TITAN input schema: 768 named features (`titan_000`–`titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.

- Tissue-source-site metadata: every row includes the grouped internal metric, random-minus-grouped change, analysed-site count, threshold-retention status, fold-size range and inner-site-separation audit.
- Binary reliability metadata: held-out average precision, cohort-specific PPV/NPV, outer and inner class-count minima, selected-component distribution and repeat-to-repeat prediction stability.
- Default model selection: binary models require at least 50 TCGA participants in each class; 17 smaller-class models are available only by explicit opt-in.
- Scope: TCGA tissue-source-site-grouped internal validation is not institutional or scanner-level external validation.
- Intended use: research-software demonstration only; TCGA benchmark without independent external validation.

Endpoint interpretation and target sources

For continuous endpoints, the TCGA OOF prediction percentile (not probability) locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, every TCGA OOF score rank (not probability) is the relative position of an uncalibrated LDA score. It is not confidence or a clinical-risk estimate. PR-AUC is non-interpolated average precision for positive class 1. PPV and NPV are held-out estimates at the observed endpoint prevalence in this TCGA analysis and will change when prevalence changes. Rows marked exploratory/limited evidence are excluded from default inference and require explicit opt-in.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **TP53 [driver_mutation].** Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **MSI-H (MANTIS >0.4).** Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.
- **MSI-H strict (MANTIS >0.6).** Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.
- **MYC.** Binary prediction that the MYC oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **PI3K.** Binary prediction that the PI3K oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **TP53 [oncogenic_pathway].** Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score.** Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.

- **Deleted arm count.** Number of chromosome arms classified as deleted; larger values indicate a greater arm-level deletion burden. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score.** Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale:* source endpoint units *Model-target source:* Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score.** Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale:* source endpoint units *Model-target source:* Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **Aneuploidy Score.** Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Lymphocyte Infiltration Signature Score.** Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Nonsilent Mutation Rate.** Rate of coding mutations that alter the encoded protein. TITANPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* $\log_{1p}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **SNV Neoantigens.** Predicted neoantigen burden arising from single-nucleotide variants. TITANPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* $\log_{1p}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Silent Mutation Rate.** Rate of coding mutations that do not change the encoded amino acid. TITANPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* $\log_{1p}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction.** Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.